Model Development and Optimization Workflow

TRAINING
2015–2021
78 observations

LEAKAGE-FREE FORECASTING DATASET

126 monthly observations | 59 candidate predictors

TRAINING-ONLY FEATURE RANKING

78 training observations | Absolute Pearson correlation

VALIDATION-BASED FEATURE SELECTION

7 candidate feature-set sizes: 5, 10, 15, 20, 30, 40, 50
24 validation observations | Ridge α = 1.0

VALIDATION
2022–2023

OPTIMAL FEATURE SET

15 predictors selected | Validation RMSE = 1.3234

RIDGE HYPERPARAMETER OPTIMIZATION

11 candidate α values evaluated
15 predictors held fixed | Validation-only tuning

FROZEN FINAL MODEL SPECIFICATION

Ridge Regression | 15 predictors | α = 0.05
Validation RMSE = 1.3150

INDEPENDENT MODEL EVALUATION

24 unseen observations | January 2024 – December 2025
No test data used during model selection or tuning

TEST SET
2024–2025

Feature ranking and model fitting used training data only; validation data guided model selection.